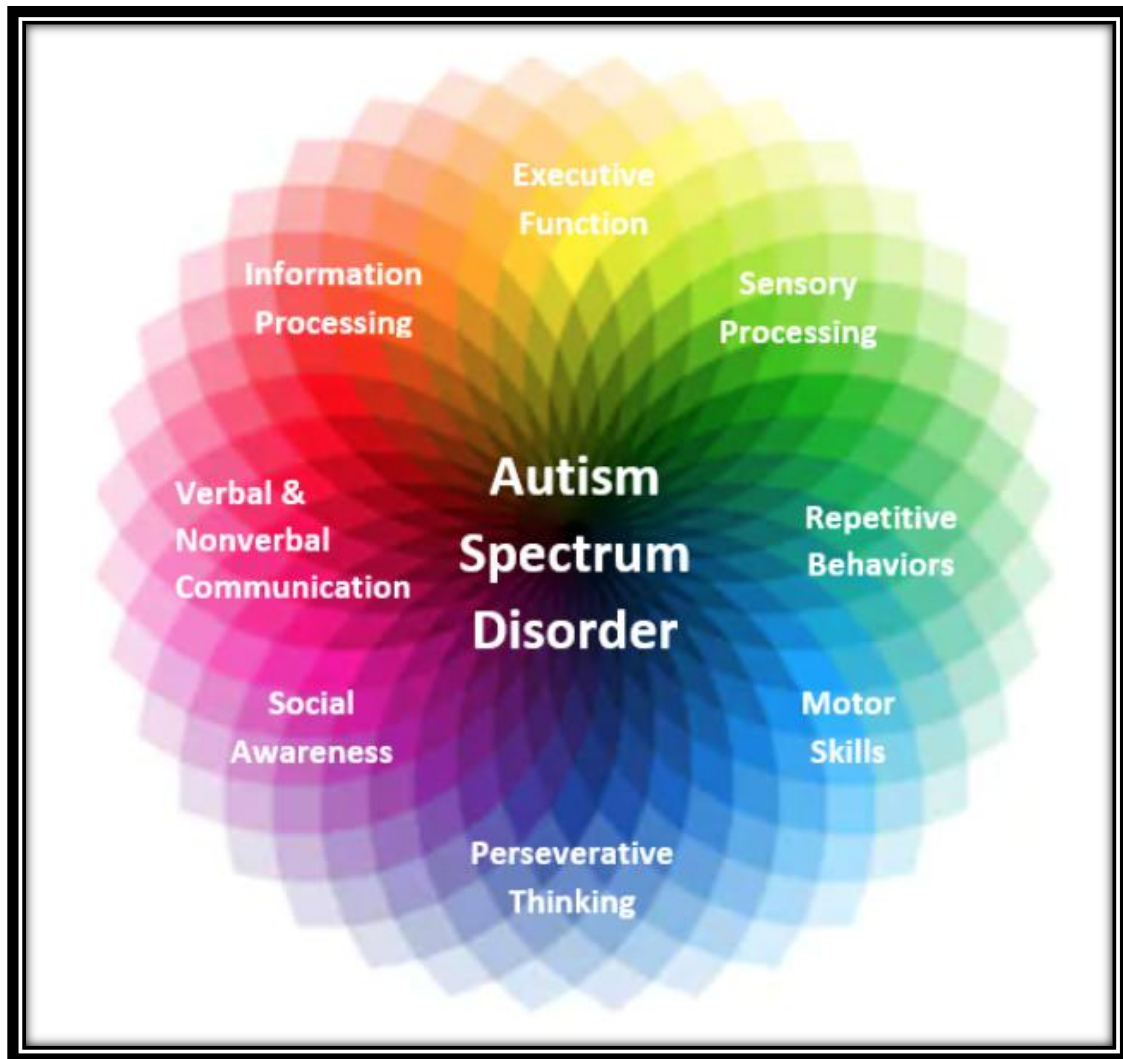


A Data-Driven Approach to Supporting Autism Screening in Adults



Jasmine (Jamie) Baldwin

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Statistics 6302: Project

Introduction

Background and Context

Autism spectrum disorder (ASD) diagnoses in adults are often delayed due to limited screening resources and lengthy assessment protocols, which can take several hours over multiple sessions and require long waitlists. This delay may prevent timely access to support services or accommodations, particularly for individuals who may have masked or gone undiagnosed during childhood. The total time from seeking an assessment to receiving a diagnosis can stretch over several months. (Ally Pediatric Therapy)

In response to these challenges, there is growing interest in using data-driven methods to improve the efficiency of preliminary autism screening. While statistical models cannot and should not replace a formal diagnosis by a licensed clinician, they can serve as tools to flag individuals who may benefit from a full diagnostic evaluation, thereby supporting a more streamlined pathway to care.

The dataset used for this analysis is titled, *“Autistic Spectrum Disorder Screening Data for Adults”* and contains 704 observations and 21 variables. It includes 10 binary responses to screening survey items from the Autism Spectrum Quotient 10 (AQ-10), a total screening score, demographic data (such as age and gender), and several medical history indicators (including whether the individual was born with jaundice) and the outcome variable of if the subject is autistic or not (Yes/No). The data contains mixed types (e.g., numerical, Boolean, and categorical variables) and includes missing values, which will be addressed during preprocessing. A comprehensive list of the variables are in Appendix A.

Before continuing here is some additional information about the AQ-10 that will help in understanding how the prompts are scored and what further exploratory data analysis of the variables distributions could mean. The AQ-10 is a short survey (2 – 5 minutes) of 10 prompts where the respondent is to answer “Definitely Agree”, “Slightly Agree”, “Slightly Disagree”, or “Definitely Disagree” to each prompt. The test is designed for adults (age 16+) with suspected autism who do not have a learning disability. Note: it makes no difference to the score whether the subject chooses “Slightly” or “Definitely”. Only 1 point can be scored for each question, meaning the max score for the survey is a 10. The threshold score is 6, meaning if the subject gains an overall sum of 6 or greater the survey indicates that they may be autistic. However this is not a definitive assessment and the results should be taken with caution as around 80% of people diagnosed with ASD score a overall sum of 6 or higher. (Embrace Autism)

To view the exact prompts asked and which result in a plus 1 to the subjects overall score, refer to appendix B.

Project Goals

The central research question guiding this project is: **Can a logistic regression model accurately predict the likelihood of an autism diagnosis in adults based on demographic factors and survey responses?**

To address this question, the analysis will proceed in several stages. First, exploratory data analysis will be conducted to better understand the distribution and relationship among variables. Second, pre-processing the data set, including but not limited to imputation, data cleaning, and up-sampling.

Ultimately, the goal is not to create a definitive diagnostic tool, but rather to demonstrate how classical statistical methods might support faster and more accessible autism screening processes. The results

may offer insight to which survey items or demographic factors are most informative and inspire future efforts to integrate data science into clinical screening workflows.

Statistical Methods

Exploratory Data Analysis

Our dataset contains 10 variables that are the subjects survey question scores for prompts 1 through 10, the summed score of the survey question scores, if they have used the app for scoring previously, if they were born with jaundice, the relation of the survey respondent to the subject, as well as the following demographic variables, age, gender, ethnicity, country of residence, and if the screening app has classified the subject as autistic. With the outcome variable being if the subject has been diagnosed with autistic or not.

Exploration of each variable and visual depictions of the distribution took place in order to view how the variable might interact with the model. As well as if variable removal and imputation were necessary on the basis of no variance, mis-inputs, and missing values. The class imbalance of the model was also explored along with correlations and associations of the variables to the outcome variable. Lastly, possible interaction terms were explored, both through statistical significance and visual depictions.

Analysis

Base Model

To establish a foundation for modeling the likelihood of an individual being autistic, we developed a logistic regression model using a traditional statistical approach. This model predicts the probability that an individual is classified as “Yes” (autistic) based on a set of behavioral and demographic predictors.

All relevant variables were incorporated into the model, including the ten binary indicators representing survey responses, as well as age, gender, ethnicity, history of jaundice at birth, country of residence, the role of the person completing the survey, and whether the individual had previously received an autism classification from the app. To ensure the model could handle categorical data appropriately, all non-numeric predictor variables were converted into binary indicator variables. The model was then trained using a logistic regression algorithm, chosen for its interpretability and suitability for binary classification tasks.

To assess the generalizability of the model, we applied repeated 10-fold cross-validation, repeating the process five times. This resampling strategy partitions the dataset into training and validation subsets multiple times, ensuring a robust and stable estimate of model performance. Importantly, the resampling procedure was stratified by the outcome variable to maintain the proportion of autistic vs. non-autistic classifications in each fold.

Model performance was evaluated using a variety of classification metrics, including accuracy, the area under the ROC curve (AUC), sensitivity (true positive rate), specificity (true negative rate), Cohen’s Kappa (chance-adjusted agreement), and the Brier score (probabilistic prediction error). These measures collectively provided a comprehensive understanding of how well the base logistic regression model performed in predicting autism classification. After evaluation, the model was refit on the full dataset to

examine the direction and magnitude of predictor effects. Coefficient estimates were interpreted to identify which variables were associated with higher or lower odds of being classified as autistic.

Elastic Net Model

The Elastic Net Model extends the base logistic regression model by incorporating elastic net regularization and addresses the class imbalance. To mitigate issues related to class imbalance, the model applies up-sampling during preprocessing, ensuring the minority class (autistic individuals) is equally represented in the training data. This helps improve the model's ability to detect autism cases, particularly in metrics sensitive to class imbalance such as sensitivity and Cohen's kappa. All other aspects, including model type, cross-validation procedure, and performance metrics, remain consistent with the base model.

Custom Model 1

Custom Model 1 extends the base logistic regression by incorporating two interaction terms and addressing class imbalance through up-sampling. Specifically, the model includes interaction effects between ethnicity and screening result, and between gender and jaundice. These interactions allow the model to account for situations where the relationship between a predictor and autism status may differ depending on another variable's value.

To mitigate issues related to class imbalance, the model applies up-sampling during preprocessing, ensuring the minority class (autistic individuals) is equally represented in the training data. This helps improve the model's ability to detect autism cases, particularly in metrics sensitive to class imbalance such as sensitivity and Cohen's kappa.

All other aspects, including model type, cross-validation procedure, and performance metrics, remain consistent with the base model.

Custom Model 2

Custom Model 2 builds on the previous setup by using a penalized logistic regression model that includes elastic net regularization. This approach helps prevent overfitting and deals with multicollinearity by combining both lasso and ridge penalties. Instead of setting the strength and type of regularization ahead of time, both were tuned during model training to find the combination that worked best for the data.

To search for good tuning values, a space-filling design was used to cover a range of options efficiently. Once the best settings were found (based on performance), the final model was fitted and evaluated again. As part of this process, only the predictors with non-zero coefficients were kept at the end, highlighting which variables the model actually relied on after applying regularization.

Results

Exploratory Data Analysis

Variable Types & Distributions

Survey Prompt Scores

Figure 1, displays the distributions of our subjects answers to survey prompts 1 through 10.

For prompt 1 we can see that the majority of subjects agreed with the prompt: “I often notice small sounds when others do not”. Which would be scored as a plus 1, indicating that this may not be a good predictor as the majority of the dataset are allistic (not autistic).

The prompt 2 - 5 distributions show a minor to a almost non-existent split between the two responses, indicating that they may not be strong predictors.

Prompt 6's distributions shows that over 1/3 of the subjects disagreed with the prompt: “I know how to tell if someone listening to me is getting bored”. Disagreeing with this statement will result in a plus 1 to the overall score. Since the majority of subject agree with this statement (and this no points are added), this could indicate that the response to this prompt is a good predictor.

For prompt 7 we see that the majority of subjects disagree with the prompt: “When I’m reading a story I find it difficult to work out the characters’ intentions”. Which results in no points added to the sum, indicating that this may a good predictor.

The distribution for prompt 8 shows that the majority of the subjects agree with the prompt: “I like to collect information about categories of things (e.g. types of car, types of bird, types of train, types of plane etc)”. Resulting in a plus 1 to the overall sum, indicating that this may not be a good predictor.

The majority of subjects agree with prompt 9: “I find it easy to work out what someone is thinking or feeling just by looking at their face”. Resulting with no points added to the sum, indicating that this may be a good predictor.

As for prompt 10: “I find it difficult to work out people’s intentions”, the majority of subjects agree with it indicating that it may not be a good predictor.

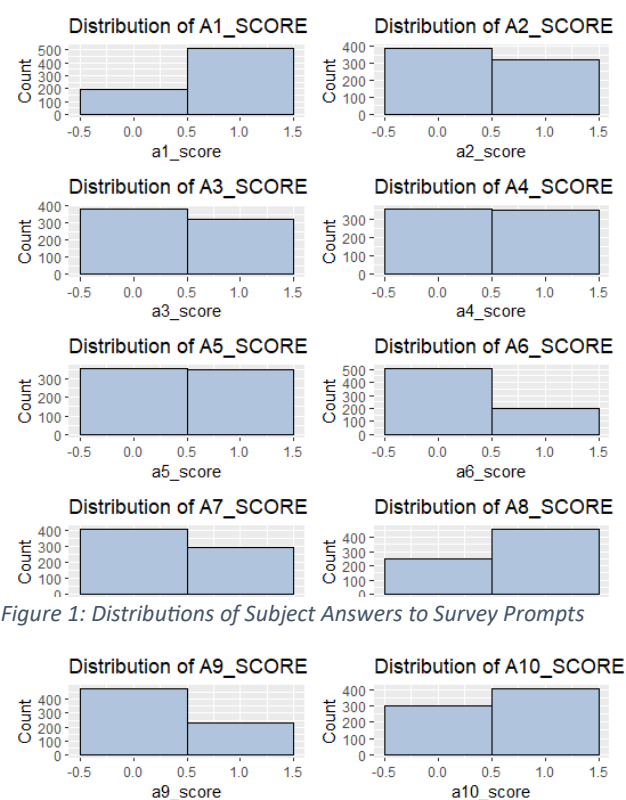


Figure 1: Distributions of Subject Answers to Survey Prompts

Total Score/Sum of Score

While our distribution located in figure 2 is not symmetrical it is moderately normal that the majority of respondents' overall scores being 4 or below. This suggests that the majority of respondents are allistic which reflects the truth discovered from the no-information rate.

Age

Figure 3 shows the distribution of age before we cleaned our data and we can see that there is a value that is certainly been incorrectly recorded as having a subject age be 383 is implausible. To rectify this the assumption that the intended age was 38 was made, as the ages in the dataset range from 17 to 64. After replacing 383 with 38, the new distribution is displayed in figure 4. Age seems to be right skewed suggesting most respondents are younger adults. We may consider transforming or binning age if needed.

Gender

There seems to be an even amount of respondents that are female and male as the distribution in figure 5 shows a near symmetrical split between the two genders. This is good for the model's generalizability across genders.

Ethnicity

The distribution of ethnicity displayed in figure 6 shows the majority of respondents being of White-European descent which may affect model generalizability across the ethnicities. As different cultures may have behavior that are more acceptable or less acceptable that may have autistic characteristics front differently when compared to other ethnicity groups.

Top 10 Countries of Residence

From figure 7 we can see that nearly 120 subjects reign from the United States then followed by the United Arab Emirates, New Zealand, India, and the UK with roughly 75 subjects reigning form each country. The countries listed afterwards continually lessen in amount of subjects reigning for them. However I feel like this is a nice enough mix of countries as only roughly 1/7 reign from solely the United States, so this may prove to be helpful to the models generalizability.

Jaundice at Birth

The distribution of subjects diagnosed with jaundice at birth is imbalanced with roughly 650 not having jaundice meaning roughly only 50 were (Fig. 8). This could potentially be a good predictor as some studies have shown that there was a statistically significant association between neonatal jaundice and ASD. (Amin SB, Smith T, Wang H.)

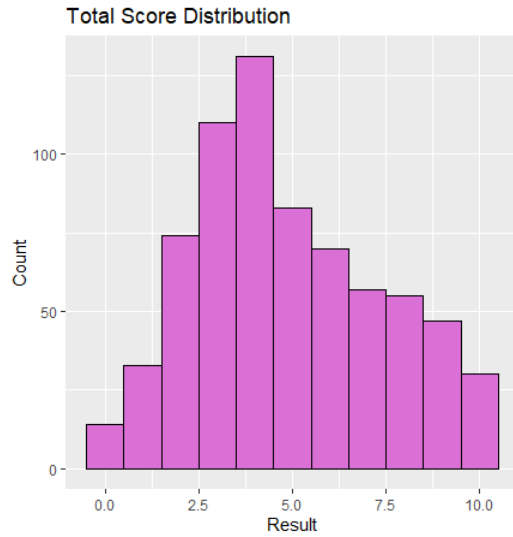


Figure 2: Distribution of Subject's Total Score

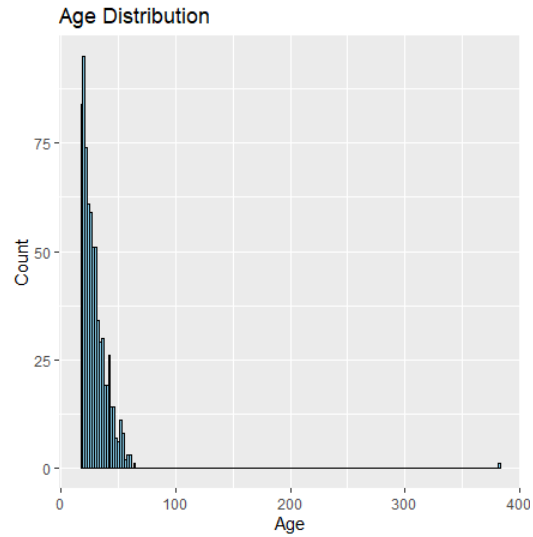


Figure 3: Distribution of Age before cleaning

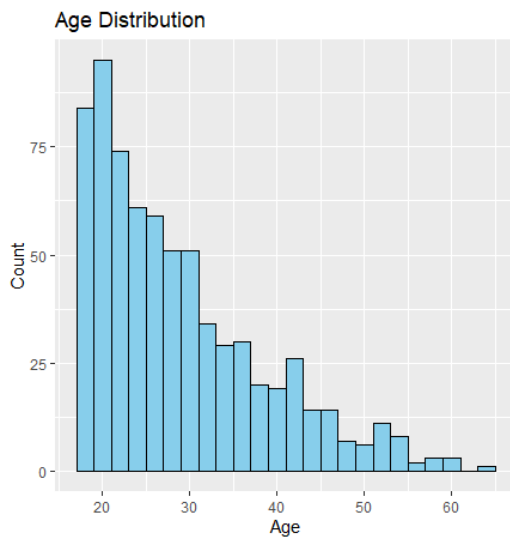


Figure 4: Distribution of Subject's Age after cleaning

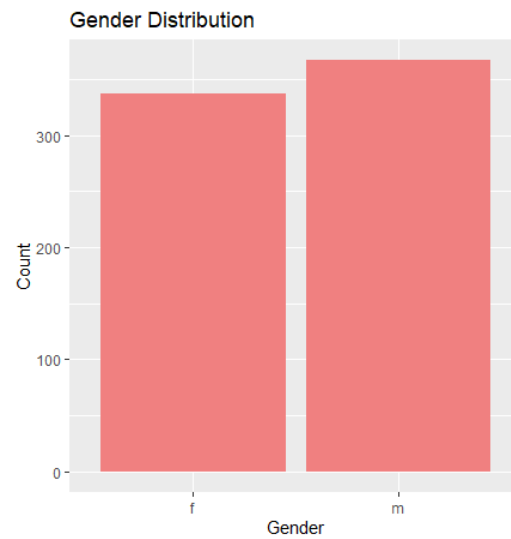


Figure 5: Distribution of Subject's Gender

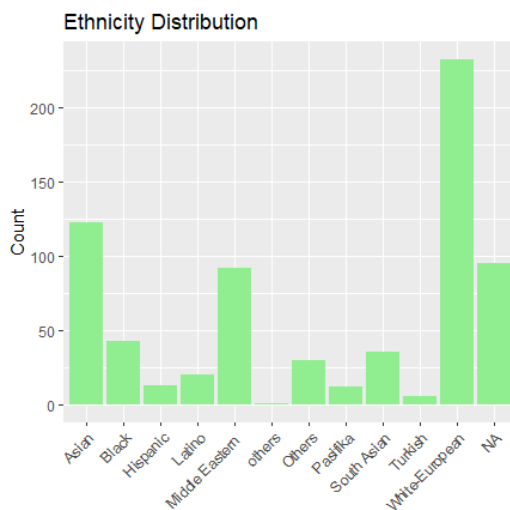


Figure 6: Distribution of Subject's Ethnicity

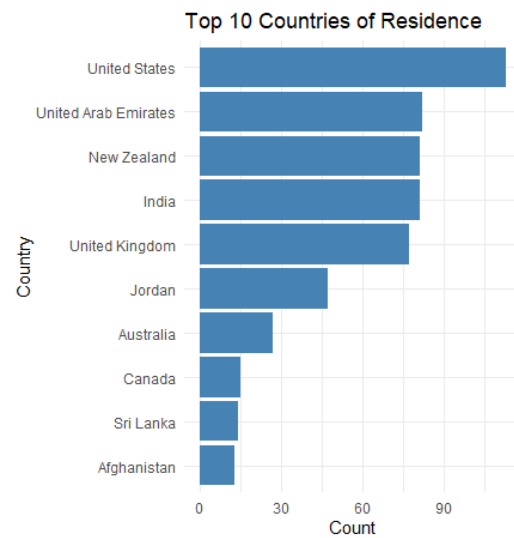


Figure 7: Distribution of Subject's Country (Top 10)

Survey Respondent Relation

Based on figure 9 the majority of subjects responded to the survey themselves, with the next highest category being NA but the next highest person to fill the survey being the subject's parent. This could suggest that the majority of those whom are autistic and filling out the survey are low-support needs with some potentially having higher support needs.

App Usage History

The distribution shown in figure 10 tells us that nearly all 700 subjects have not used this phone app before to screen themselves for autism. However, this does not mean they have never been screened for autism before. This may not be necessary as a predictor for this model as this was data pertaining to the report this data was originally gathered for.

Survey Classification

Figure 11 holds the comparison of the distribution of the AQ-10 classifications and the true diagnosis distribution. It depicts how the classifications from the AQ-10 and the true diagnosis are similar in their proportions. However, there is indications that the survey tends to over flag allistic individuals as autistic.

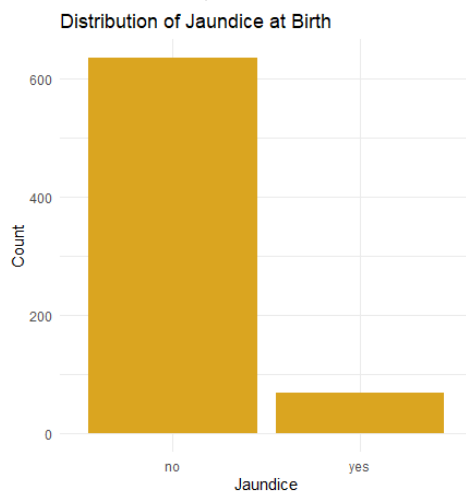


Figure 8: Distribution of Subject's Neonatal Jaundice

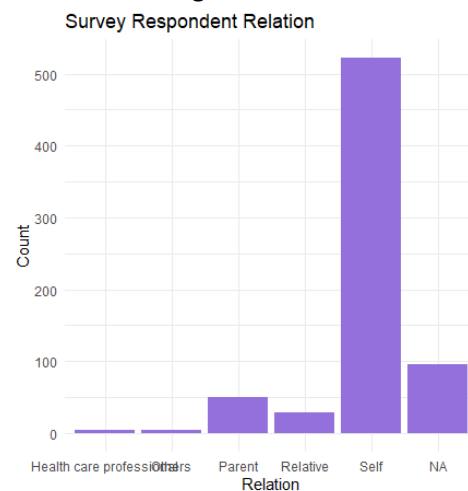


Figure 9: Distribution of Respondent Relation to the Subject

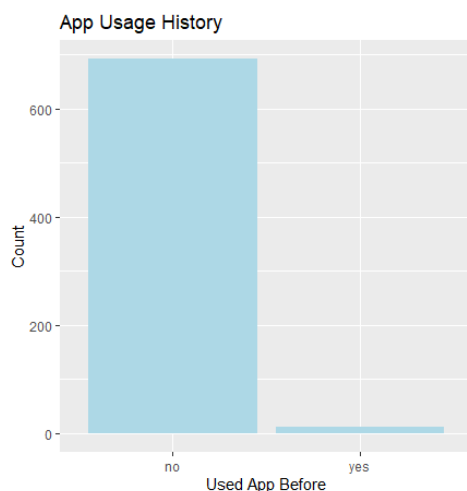


Figure 10: Distribution of Subject's App Usage History

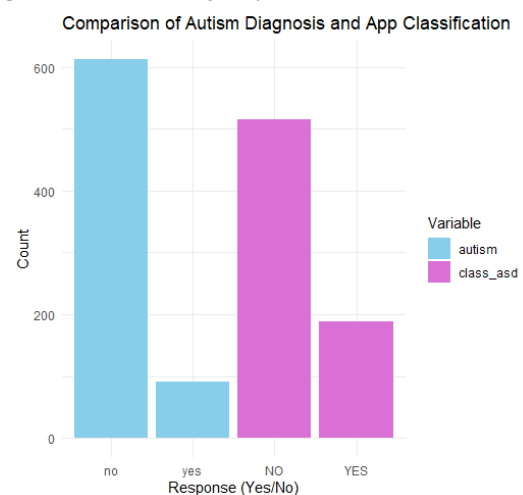
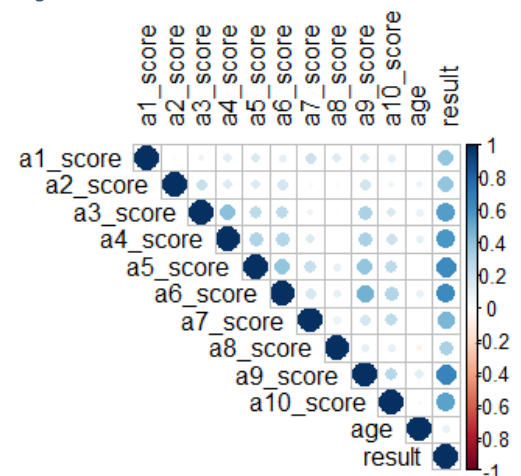


Figure 11: Distribution of Subject's App Classification Compared to the Outcome Variable



Stacked bar plots were used to explore how the proportion of individuals with autism varied across levels of categorical predictors. Differences in proportional distributions (e.g., higher autism rates in those with a history of jaundice) suggest these variables may be associated with autism status. Figure 13 contains the following variables gender of subject, relation of respondent to subject, jaundice present at birth of subject, and if the survey classified the subject as autistic or not. This figure shows that there are more females diagnosed with autism than males in our dataset. It seems that subjects that filled out the survey themselves have the lowest proportion of autism diagnosis, viewing this it might be in the best interest of the model to bin the values “Health care professional”, “Others”, “Parent”, and “Relative” into a “Not-Self” category. By doing so we would see that the subject not filling out the survey themselves may be a good predictor on if the individual may be autistic. We are also shown that the majority of autistic individuals had jaundice at birth, indicating that this may be a good predictor. The last variable in the figure is if the app/survey classified the individual as autistic or not based on their overall survey score. The plot shows that although it caught the majority of autistic individuals it did underdiagnosis some and overdiagnosis many allistic individuals.

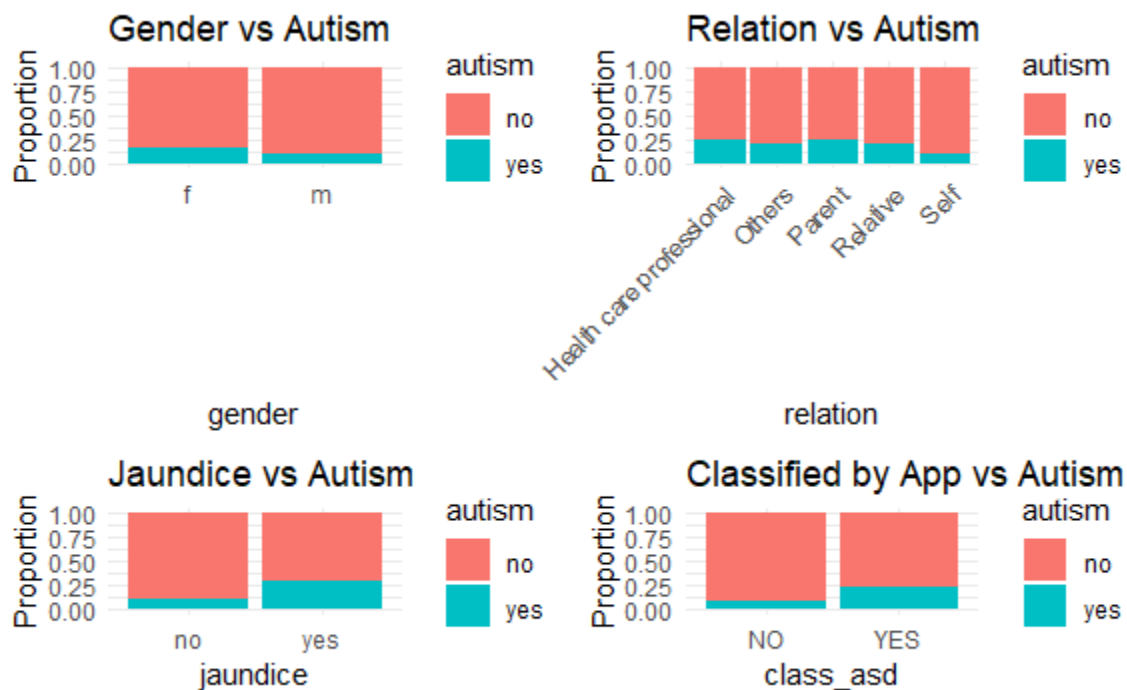


Figure 13: Gender, Relation, Jaundice, and App Classification vs Autism Diagnosis

Figure 14 displays the proportion of autistic individuals in each ethnicity group and for the top 10 countries. This ethnicity plots shows that the majority of autistic people are of Latino descent, then followed by a 3-way tie between Pasifika, Turkish, and White-European individuals. For our top 10 countries the highest proportion of autistic people reign from Afghanistan followed by, Australia, Canada, the UK, then the US. Since there is not an overwhelming majority of autistic people

concentrated in one country this may not be a good predictor for autism.

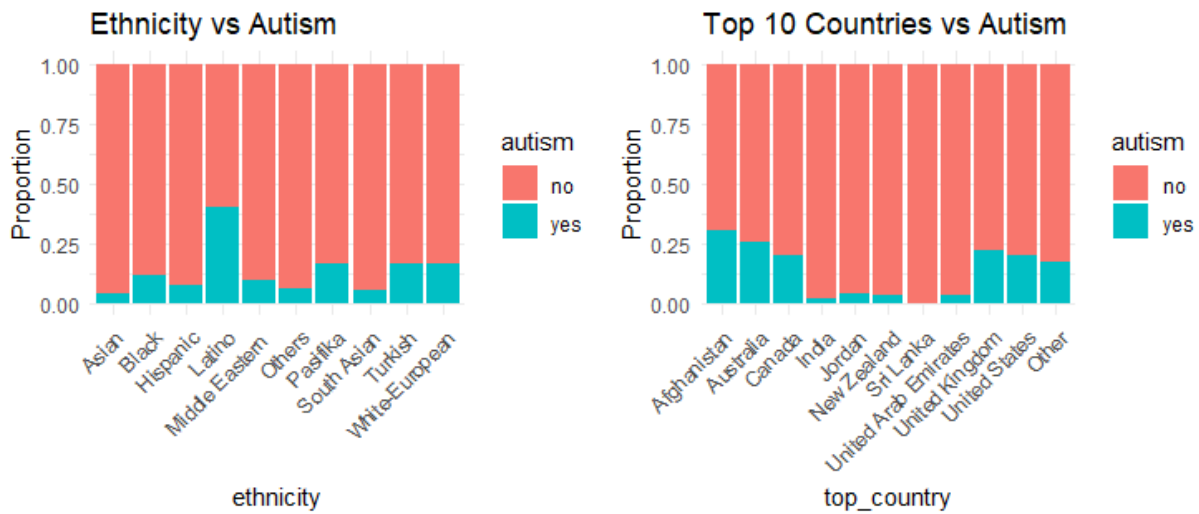


Figure 14: Ethnicity and Countries (Top 10) vs Autism Diagnosis

Interaction Terms

To identify potential interaction effects on the likelihood of an autism diagnosis, a systematic approach was employed. For every unique pair of predictor variables two logistic regression models were compared: one including only the main effects of the two predictors, and another including their interaction term. A likelihood ratio test was then performed to assess whether the model with the interaction term provided a significantly better fit to the data compared to the model with only main effects, using a significance level of $\alpha=0.05$.



Figure 15: Visualizations of Potential Interaction Terms

Pairs of predictor variables for which the likelihood ratio test yielded a p-value below this threshold were considered to have a statistically significant interaction. These significant interactions were then listed and ordered by their p-value. Following this statistical identification, visualizations of these potential

interaction terms were generated and presented in Figure 15. These plots were examined to understand the nature of the potential interactions.

Based on the visual inspection of these interaction plots in Figure 15, two interactions were prioritized for inclusion in modeling:

The interaction between ethnicity and the result of the screening questionnaire. The box plots in Figure 15 visualizing this interaction suggested that the relationship between an individual's ethnicity and their screening questionnaire result differed depending on whether they received an autism diagnosis.

The interaction between gender and history of jaundice. The stacked bar charts in Figure 15 for this interaction indicated that the proportion of individuals diagnosed with autism within each gender varied based on whether they had a history of jaundice, suggesting that the association between jaundice and autism diagnosis might be different for males and females.

The other statistically significant interactions identified by the likelihood ratio tests were not prioritized for immediate inclusion based on the visual inspection of their interaction plots in Figure 15. While statistically significant, the visual patterns in those plots did not as clearly suggest a strong moderating effect of one variable on the relationship between the other and the autism diagnosis, compared to the two interactions. Therefore, the modeling efforts were initially focused on these two interactions that showed both statistical significance and a visually interpretable potential for a meaningful interaction.

Analysis

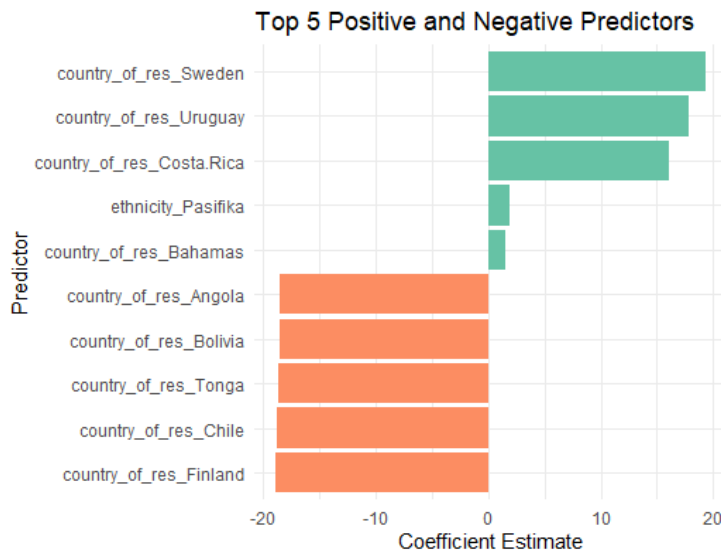
Model Comparison: Performance Metrics						
Evaluating classification models on Autism Screening						
Model	Accuracy	Brier Score	Kappa	ROC AUC	Specificity	Sensitivity
Base Model	0.846	0.124	0.116	0.620	0.951	0.139
Custom Model 1	0.696	0.219	0.156	0.608	0.721	0.528
Custom Model 2	0.683	0.201	0.183	0.711	0.692	0.625
EN Model	0.685	0.203	0.190	0.709	0.693	0.634

Table 1: Comparison of Model Metrics

Base Model

The base logistic regression model retained all 94 dummy-coded predictors, including all original survey and demographic variables. It achieved high specificity (95.1%), indicating strong performance in identifying non-autistic individuals. However, sensitivity was low (13.9%), suggesting the model struggled to correctly classify autistic cases. Overall accuracy was 84.6%, but the modest ROC AUC of 0.62 and low Cohen's kappa (0.12) indicate limited discriminative ability and poor agreement beyond chance. The Brier score of 0.124 suggests moderate calibration in probabilistic predictions.

Figure 16: Base Model's Top 5 Positive & Negative Predictors



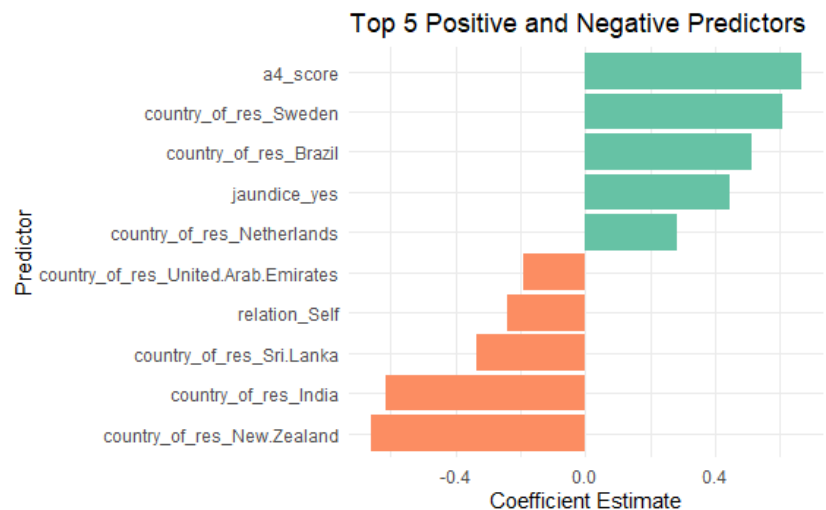
In total, the model retained 94 predictors after dummy encoding. Among these, 66 had negative coefficients while only 27 had positive coefficients, indicating the model was more frequently associating features with a decreased likelihood of autism classification. The top five positive predictors—those most strongly associated with an increased probability of being classified as autistic, were residence in Sweden ($\beta = 19.33$), Uruguay ($\beta = 17.83$), Costa Rica ($\beta = 16.04$), identifying as Pasifika ($\beta = 1.96$), and residence in the Bahamas ($\beta = 1.53$).

In contrast, the top five negative predictors, that is those most strongly associated with a

decreased likelihood of classification, were residence in Angola ($\beta = -18.48$), Bolivia ($\beta = -18.49$), Tonga ($\beta = -18.63$), Chile ($\beta = -18.79$), and Finland ($\beta = -18.88$), as seen in figure 16. These large-magnitude coefficients, particularly for country and ethnicity, suggest strong modeled effects, though they may also reflect small sample sizes or imbalanced representation in the dataset.

Beyond the extremes, additional positive predictors of interest included identifying as Latino, Turkish, or Middle Eastern, reporting jaundice at birth, and higher values on certain survey items. These more moderate effects help ground the model in both demographic and symptom-based signals.

Figure 17: Elastic Net Model's Top 5 Positive & Negative Predictors



Elastic Net Model

The base elastic net model showed fairly balanced performance, with an accuracy just under 68.1% and an AUC of about 0.71. Specificity and sensitivity were also close, around 68.6% and 64.0%, indicating the model did a decent job identifying both autistic and non-autistic cases. The Brier score and Kappa suggested moderate predictive reliability and agreement beyond chance.

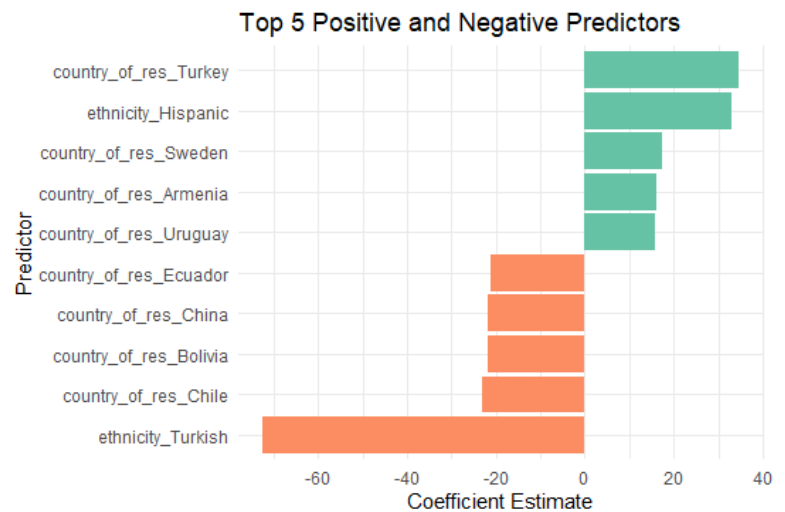
In terms of predictors, one of the strongest signals came from responses to a specific social behavior item. Country of residence played a notable role as well, with individuals from Sweden and Brazil more likely to be classified as autistic, while those from New Zealand and India were less likely. The history of neonatal jaundice also appeared as a meaningful positive factor, whereas people who filled out the survey for themselves tended to be

classified as non-autistic. These predictors are showcased in figure 17. Overall, the model leaned slightly more on positive predictors, keeping a total of 12 compared to 10 negative ones.

Figure 18: Custom Model 1's Top 5 Positive & Negative Predictors

Custom Model 1

For Custom Model 1, a total of 103 predictors were retained after dummy encoding, with 34 showing positive coefficients and 69 showing negative coefficients. This suggests that the model placed more weight on predictors associated with a decreased likelihood of being classified as autistic. The top five positive predictors, indicating the strongest associations with an increased likelihood of autism classification, were: residence in Turkey ($\beta = 34.71$), identifying as Hispanic ($\beta = 32.86$), residence in Sweden ($\beta = 17.32$), residence in Armenia ($\beta = 16.15$), and residence in Uruguay ($\beta = 15.90$). These top positive predictors are shown in Figure 18.



In contrast, the top five negative predictors, which were most strongly associated with a decreased likelihood of being classified as autistic, were: identifying as Turkish ($\beta = -72.55$), residence in Chile ($\beta = -23.25$), Bolivia ($\beta = -21.98$), China ($\beta = -21.68$), and Ecuador ($\beta = -21.15$), as depicted in Figure 18. The substantial negative coefficients suggest that these factors have a powerful inverse relationship with autism classification, but could also be influenced by factors like small subgroup sizes.

The model's relatively high specificity of 72.1% indicates it is moderately good at correctly identifying autistic cases, but the lower sensitivity (52.8%) implies it may struggle to correctly classify autistic cases on occasion.

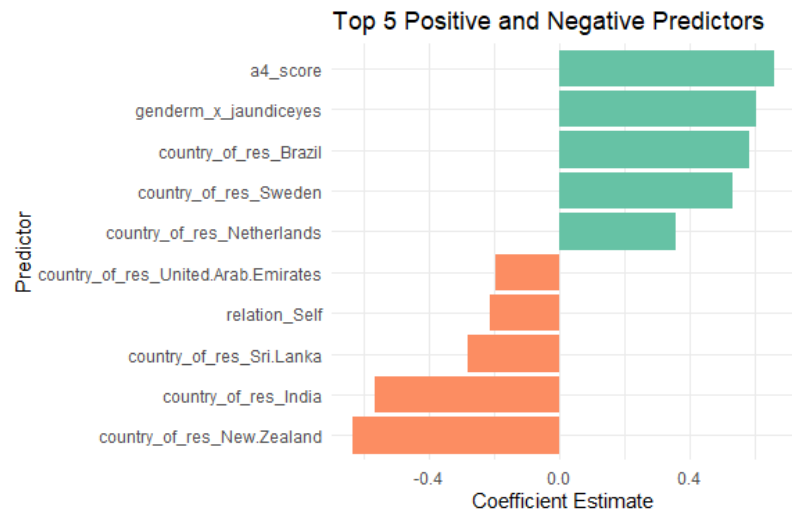
Overall, while Custom Model 1 is effective at detecting autism in individuals, its performance could benefit from addressing the imbalance in predictor types and improving its ability to differentiate non-autistic individuals more accurately. This is reflected in the model's ROC AUC score of 0.608, which shows moderate ability to distinguish between the two classes.

Custom Model 2

Custom Model 2 showed moderate classification performance overall. The model achieved an accuracy of 68.3% and an AUC of 0.71, suggesting a decent ability to distinguish between autistic and non-autistic cases. Specificity (true negative rate) was slightly stronger at 69.2%, while sensitivity (true positive rate) was lower at 62.5%, indicating the model leaned slightly toward identifying negative cases. The Brier score was 0.20, and the Kappa statistic was relatively low at 0.18, suggesting moderate agreement beyond chance.

After regularization, the model retained 23 predictors: 13 with positive coefficients and 10 with negative coefficients. The strongest positive predictors included responses to survey question 4 ($\beta = 0.658$), and being from countries like Sweden ($\beta = 0.529$), Brazil ($\beta = 0.583$), and the Netherlands ($\beta = 0.355$), as well as the interaction term of being male and having a history of jaundice at birth ($\beta = 0.605$). On the other hand, the strongest negative predictors were being from New Zealand ($\beta = -0.631$), India ($\beta = -0.562$), and Sri Lanka ($\beta = -0.279$), the respondent of the survey being the subject ($\beta = -0.211$), and being from the United Arab Emirates ($\beta = -0.196$). These predictors being featured in figure 19. Some other notable positive predictors were responses to survey question 1, being Latino, and age, though their effect sizes were smaller. Overall, the retained variables suggest that a mix of survey responses and demographic factors, including geography and neonatal jaundice, played a role in how the model classified individuals.

Figure 19: Custom Model 2's Top 5 Positive & Negative Predictors



Conclusions and Limitations

This study evaluated multiple predictive models for adult autism screening based on “self-reported” survey responses and demographic information. While all models exhibited trade-offs between sensitivity and specificity, the elastic net logistic regression model emerged as the most balanced and reliable option among those tested. It achieved reasonable accuracy (~68%), a moderate AUC (~0.708), and more equitable sensitivity and specificity compared to other models that tended to heavily favor one metric at the expense of the other. However, after viewing the models strongest variables Custom Model 2 may be the more optimal model to pick as its top 2 predictors that positively predicts autism are commonly statistically significantly associated with an autism diagnosis.

If one model must be selected for practical use, Custom Model 2 is recommended due to its relative robustness and ability to manage multicollinearity through regularization. However, this recommendation is made with significant caution. The model's moderate kappa value and calibration

shortcomings suggest that its predictions should not be interpreted as definitive indicators of autism. Instead, this model should be viewed as a preliminary screening tool, one that may help flag individuals for further evaluation, not as a diagnostic instrument.

A key limitation of this study lies in the quality and scope of the dataset. The data were “self-reported”, potentially biased, and appear to suffer from demographic imbalances, particularly in country and ethnicity variables. Some of the strongest predictors were linked to geography or demographic traits, which may reflect sampling artifacts or overfitting rather than true underlying patterns. These issues raise concerns about the generalizability of the results.

To strengthen the model's reliability and ethical use, future work must focus on acquiring more representative and diverse data, validating findings across independent samples, and assessing the impact of including demographic variables. Until such efforts are undertaken, custom model 2, and any model based on this dataset, should be used cautiously and never as a substitute for clinical assessment.

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Appendix A: Variable Descriptions

Baseline Variables

- **A1_Score** – Score of Answer 1
- **A2_Score** – Score of Answer 2
- **A3_Score** – Score of Answer 3
- **A4_Score** – Score of Answer 4
- **A5_Score** – Score of Answer 5
- **A6_Score** – Score of Answer 6
- **A7_Score** – Score of Answer 7
- **A8_Score** – Score of Answer 8
- **A9_Score** – Score of Answer 9
- **A10_Score** – Score of Answer 10
- **age** – Age of subject in years.
- **gender** – Gender/Sex of subject. Male or Female.
- **ethnicity** – Ethnicity of subject.
- **jaundice** – Whether the subject was born with jaundice.
- **autism** – Whether the subject is diagnosed with autism or not. (outcome variable)
- **country_of_res** – Subject's country of residence.
- **used_app_before** – Whether the subject has used the screening app before.
- **result** – Sum of Scores A1 – A10.
- **age_desc** – Age description of subject.
- **relation** – Whether the Subject answered themselves or if someone did on their behalf.
- **Class/ASD** – Whether the app classified them as autistic or not.

Appendix B: AQ-10 Survey Questions and Scoring

The following questions are from the Autism Spectrum Quotient. They are to be answered with one of the following: Definitely Agree (DA), Slightly Agree (SA), Slightly Disagree (SD), Definitely Disagree (DD).

Scoring: Only 1 point can be scored for each question. Score 1 point for Definitely or Slightly Agree on each of items 1, 7, 8, and 10. Score 1 point for Definitely or Slightly Disagree on each of items 2, 3, 4, 5, 6, and 9. If the individual scores 6 or above, consider referring them for a specialist diagnostic assessment.

A1_Score Question: I often notice small sounds when others do not.

A2_Score Question: I usually concentrate more on the whole picture, rather than the small details

A3_Score Question: I find it easy to do more than one thing at once

A4_Score Question: If there is an interruption, I can switch back to what I was doing very quickly

A5_Score Question: I find it easy to 'read between the lines' when someone is talking to me

A6_Score Question: I know how to tell if someone listening to me is getting bored

A7_Score Question: When I'm reading a story I find it difficult to work out the characters' intentions

A8_Score Question: I like to collect information about categories of things (e.g. types of car, types of bird, types of train, types of plant etc)

A9_Score Question: I find it easy to work out what someone is thinking or feeling just by looking at their face

A10_Score Question: I find it difficult to work out people's intentions